

Unitary equivalence of two representations

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.UnitaryEquivalent`

def

Defines equivalence by one surjective isometry intertwining all algebra actions.

Statement

Let A be a complex C^* -algebra, with no unit assumed, and let $\pi : A \rightarrow \mathcal{B}(H)$ and $\rho : A \rightarrow \mathcal{B}(K)$ be $*$ -representations on complex Hilbert spaces. They are unitarily equivalent precisely when the following condition holds.

Definition

There exists a surjective complex-linear isometry $U : H \rightarrow K$ such that

$$U(\pi(a)x) = \rho(a)(Ux) \quad \text{for every } a \in A \text{ and } x \in H.$$

Equivalently, $U\pi(a) = \rho(a)U$ for all a , or $\rho(a) = U\pi(a)U^{-1}$. The inverse exists because U is onto as well as isometric. An abstract isomorphism of H and K without these intertwining equations is not the defined condition.

Assumptions

Both representations have the same source algebra. Neither irreducibility, nonzero action, separability, nor finite dimension is required.

Conclusion

One unitary from H onto K intertwines the two actions of every algebra element.

Main citations

- Exact definition and context
- The representation interface

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
def UnitaryEquivalent (pi : NonUnitalCStarRepresentation A H)
  (rho : NonUnitalCStarRepresentation A K) : Prop :=
  ∃ U : H ≈li[ $\mathbb{C}$ ] K, ∀ (a : A) (x : H), U (pi a x) = rho a (U x)
```

In the source	Mathematical meaning
$(\pi : \text{NonUnitalCStarRepresentation } A \ H)$	The representation π on the complete complex Hilbert space H .
$(\rho : \text{NonUnitalCStarRepresentation } A \ K)$	The representation ρ of the same A on the complete complex Hilbert space K .
$\exists U : H \approx_{\text{li}}[\mathbb{C}] \ K$	Choose one complex-linear isometric equivalence $U : H \rightarrow K$, including surjectivity and an inverse.
$\forall (a : A) (x : H)$	That same U must work for every algebra element and every vector in its domain.
$U (\pi \ a \ x) = \rho \ a \ (U \ x)$	Act by $\pi(a)$ in H and then apply U , or first apply U and then act by $\rho(a)$ in K ; the two resulting vectors are equal.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.UnitaryEquivalent

Accepted content SHA-256: 098cb190191b380a9c3446b3de808b5737235cbd46c11ee75113ccf4ca622ea0

Accepted source guide SHA-256: 6a635660384b2f89fd8a4b3786b7fcf4bc221d50a0b2942732d496444b8c3e7f

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73